

Coding & Solution

Date	24 June 2026
Team ID	SWTD-2026-3249
Project Name	OptiCrop - Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of the implemented OptiCrop code and the overall solution. The submission demonstrates working functionality, clean code practices, and alignment with the defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/opticrop-team/opticrop (private repo — access available on request)
Programming Language(s)	Python, JavaScript, Dart
Framework(s) Used	Django REST Framework, Flask (ML microservice), React.js, Flutter
Key Features Implemented	- ML-based crop recommendation and yield prediction engine - Real-time, weather-aware irrigation and fertilizer advisory - Offline-first Flutter mobile app for farmers - Geospatial field mapping via Google Maps API - Basic SMS alerting via Twilio
Pending / Incomplete Features	- Multilingual SMS templates for regional languages - Automated model retraining pipeline from farmer feedback

Setup / Run Instructions	1. Clone the repository and install backend dependencies with <code>pip install -r requirements.txt</code> 2. Configure environment variables for the database, OpenWeatherMap, Twilio, and Google Maps API keys 3. Run database migrations, then start the Django server with <code>python manage.py runserver</code> 4. Install Flutter dependencies with <code>flutter pub get</code> and run the mobile app with <code>flutter run</code>
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Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The current build covers the core recommendation engine, mobile app, and geospatial mapping end-to-end. Remaining work focuses on expanding regional-language SMS support and automating the feedback-driven model retraining pipeline ahead of the next release.